Futsal
Booking and Management System

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Outline

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- Objectives
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Introduction

Our System is a web-based multi-tenant SaaS platform.

System enables users to search and book futsal courts online and forum

Allowing owners to manage bookings, schedules, pricing, and reviews through a centralized dashboard.

The logistic regression algorithm is used for sentiment analysis of user reviews.

Problem Statement

- Manual bookings such as phone calls, handwritten records cause double booking.
- Users struggle to find and compare futsal centers.
- Difficulty managing bookings, schedules, and revenue for futsal owners.
- Lack of proper analytics is leading to poor business decisions.

Objectives

- To provide a centralized platform for futsal discovery and online court booking.
- To simplify booking, scheduling and revenue management for futsal owners.
- To provide booking analytics for better decision-making.

Scope

- Search futsal centers by location, availability, and pricing
- Owner dashboard for courts, time slots, pricing, and customers.
- Community features: ratings, reviews, and discussion forums.
- Location-based recommendation engine for nearby futsals.

Limitations

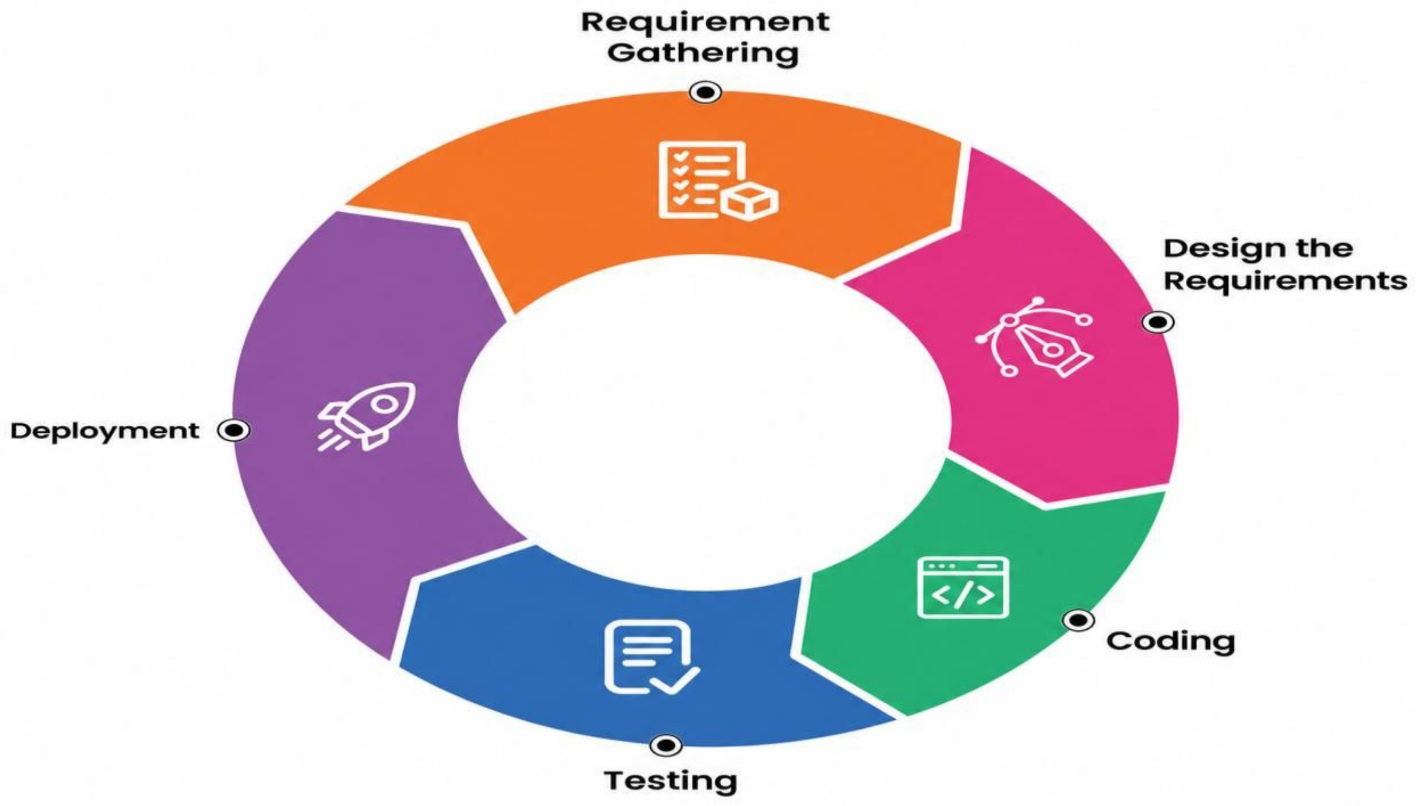
- Dependent on stable internet connectivity to function properly.
- Mobile app support is limited in this initial phase (web-first).
- Requires basic user training for owner and admin dashboards.

Literature Review

- Suhaimi & Sahid (2021) – Digital Shift Recommends replacing manual booking records with digital systems.
- Fauzi et al. (2022) – Operational Efficiency Web-based systems improve accessibility and reduce manual errors.
- Modern System Design – SaaS Architecture Multi-tenant SaaS ensures scalability and secure data isolation.
- Machine Learning Trends – AI & Analytics Logistic Regression and TF-IDF improve sentiment analysis and user insights

Development Methodology

Developed using the Agile methodology across five iterations, emphasizing flexibility, collaboration, and continuous feedback.



Requirement Analysis

Functional Requirements	Non-Functional Requirements
User Registration and Authentication	Usability-simple, clean interface fro all user roles.
Futsal discovery & search by location, price	Security-encrypted data, HTTPS, role-based access.
Booking with slot-conflict prevention	Performance- handles concurrent requests smoothly
Ratings, reviews & personalized court recommendations.	Maintainability-modular architecture, easy to update.
futsal owner dashboard: court, schedule, pricing	Reliability-accurate, consistent results every time.

Feasibility Study

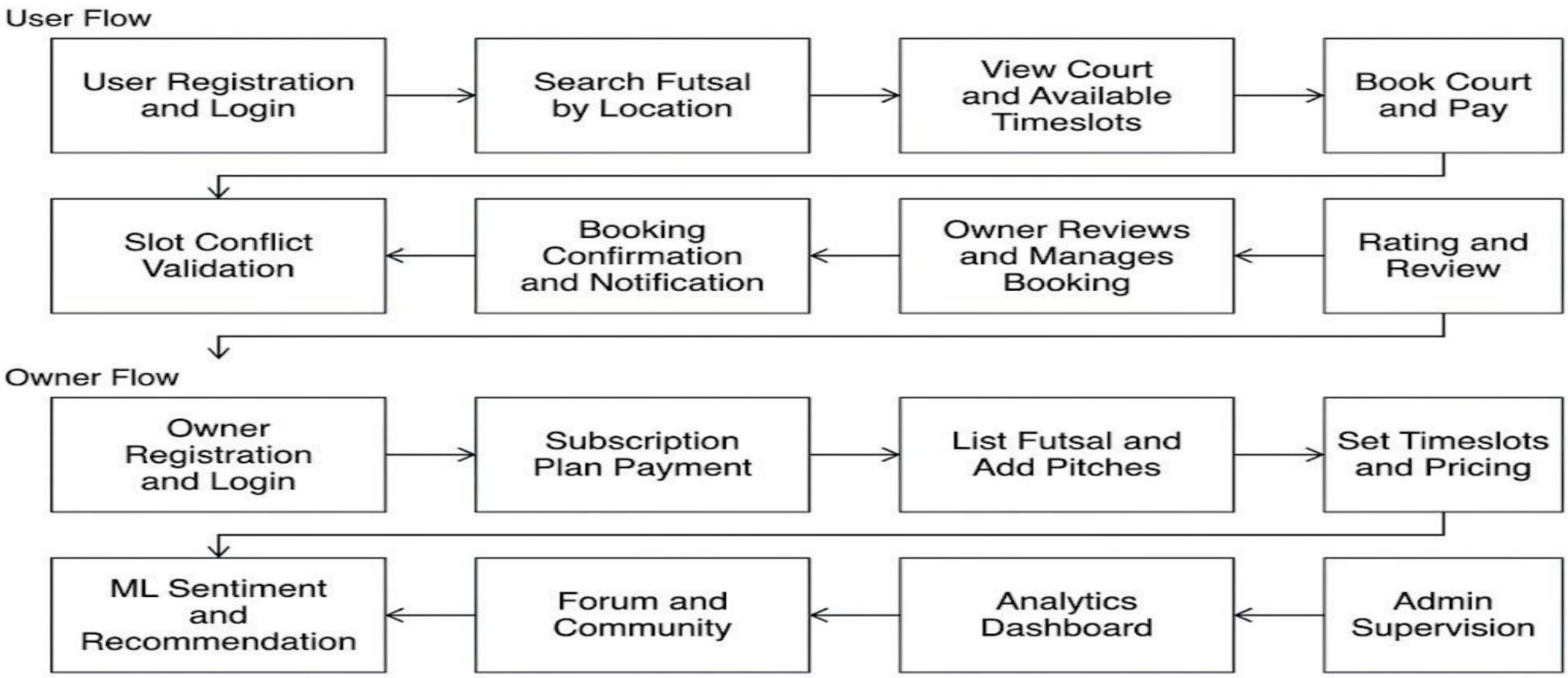
- **Technical Feasibility**
  Built using readily available and reliable technologies: React, Node, Redis, Scikit-learn, docker.

- **Economic Feasibility**
  Developed at minimal cost using open-source technologies and free-tier cloud hosting services.

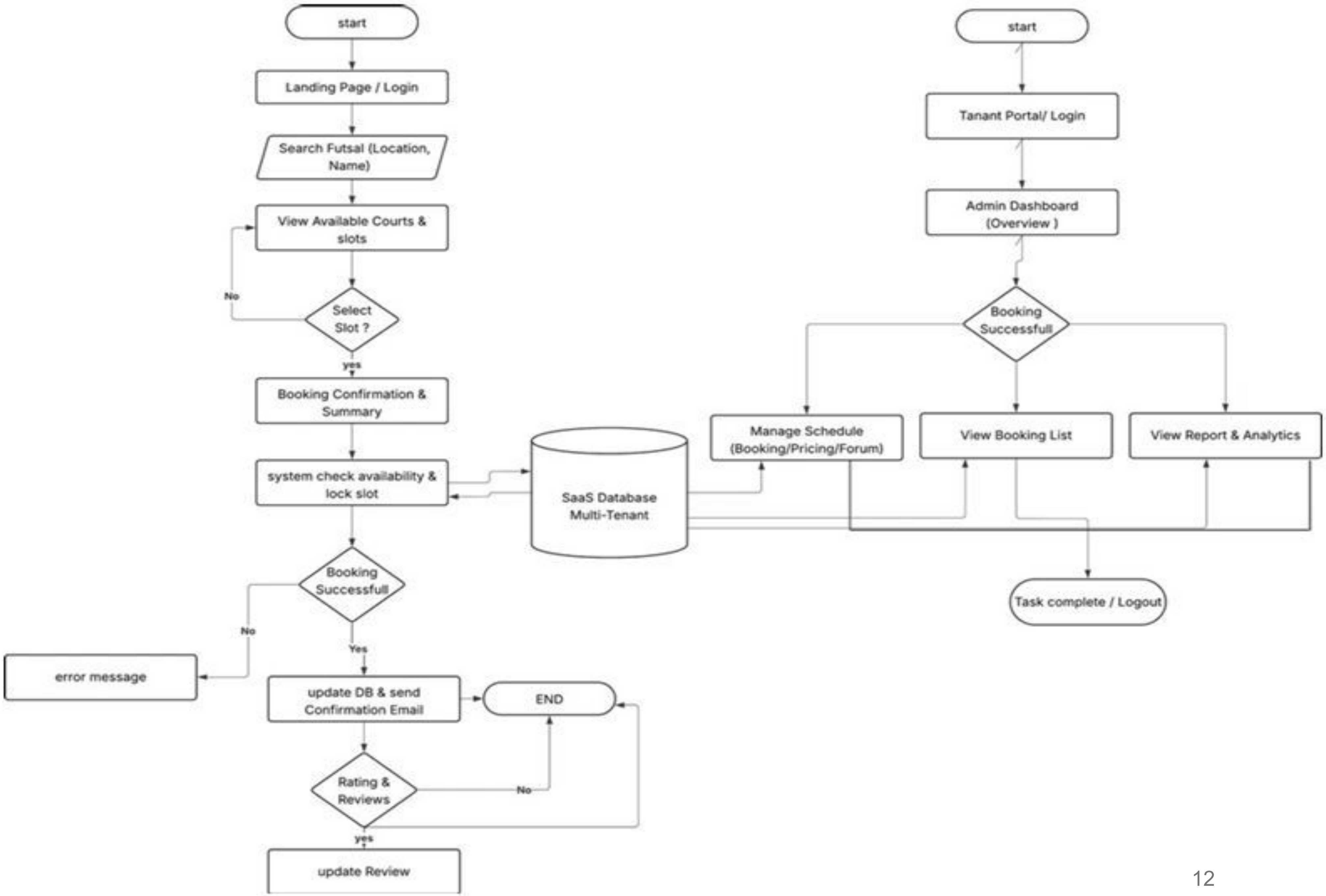
- **Operational Feasibility:**
  Browser-based with no installation. Minimal training needed; owners just register and keep availability updated.

- **Schedule Feasibility**
  Completed within 14 weeks across 10 phases, following Agile with overlapping tasks for timely delivery.

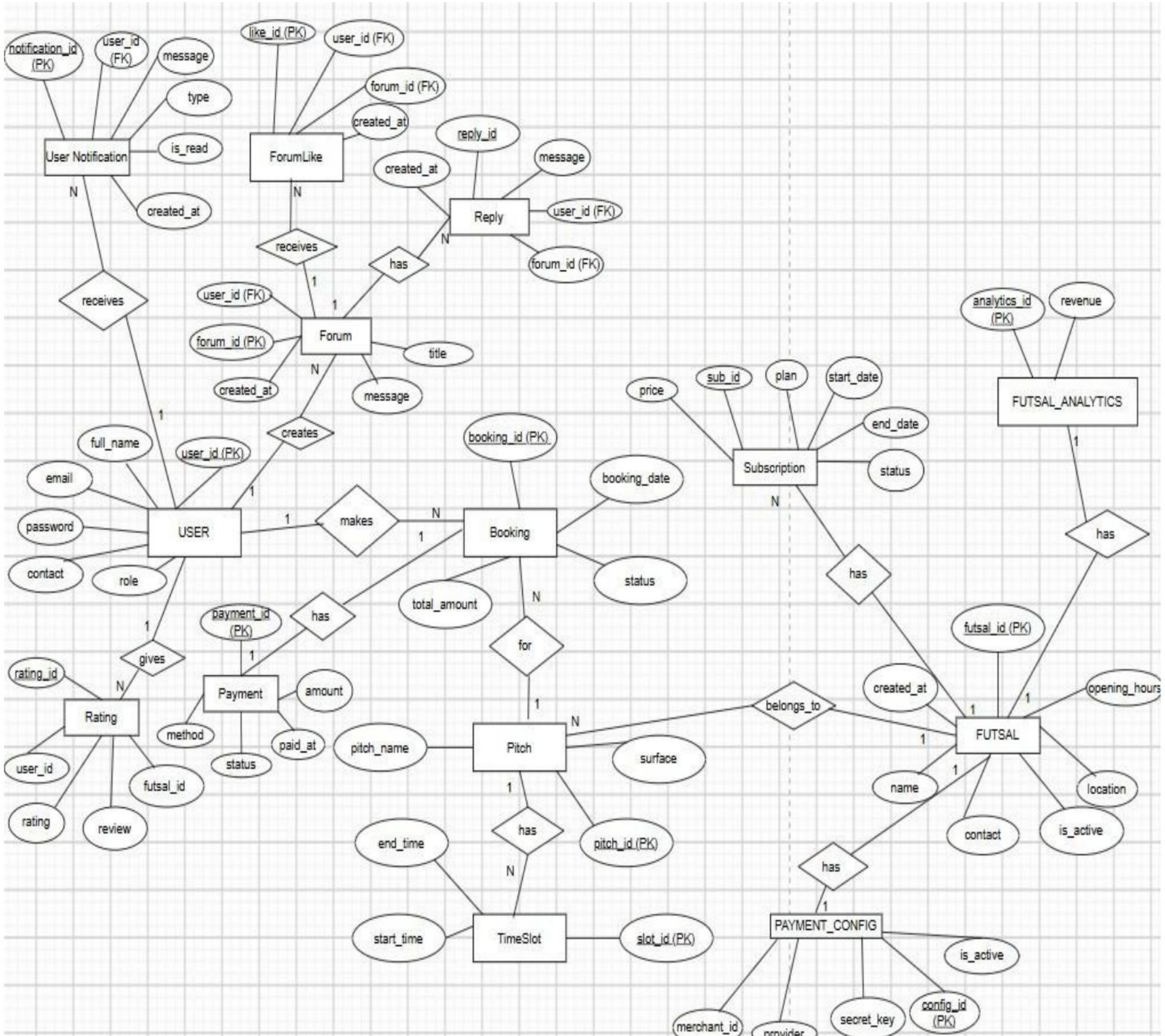
System Architecture



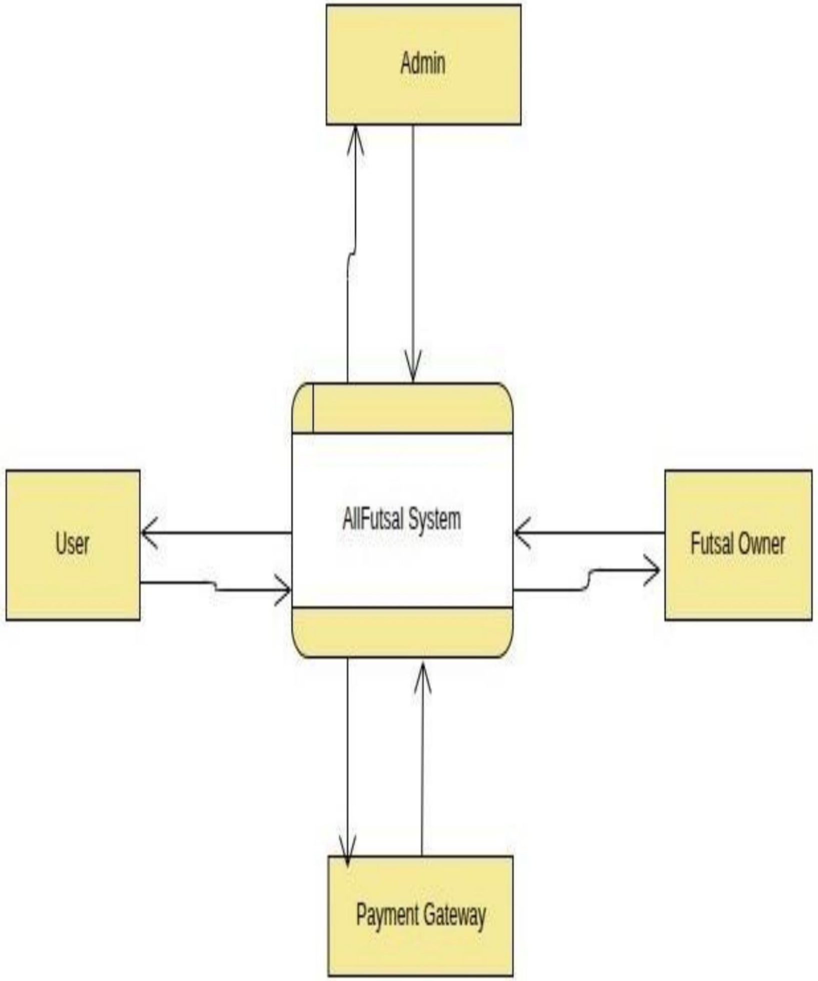
Workflow of the system



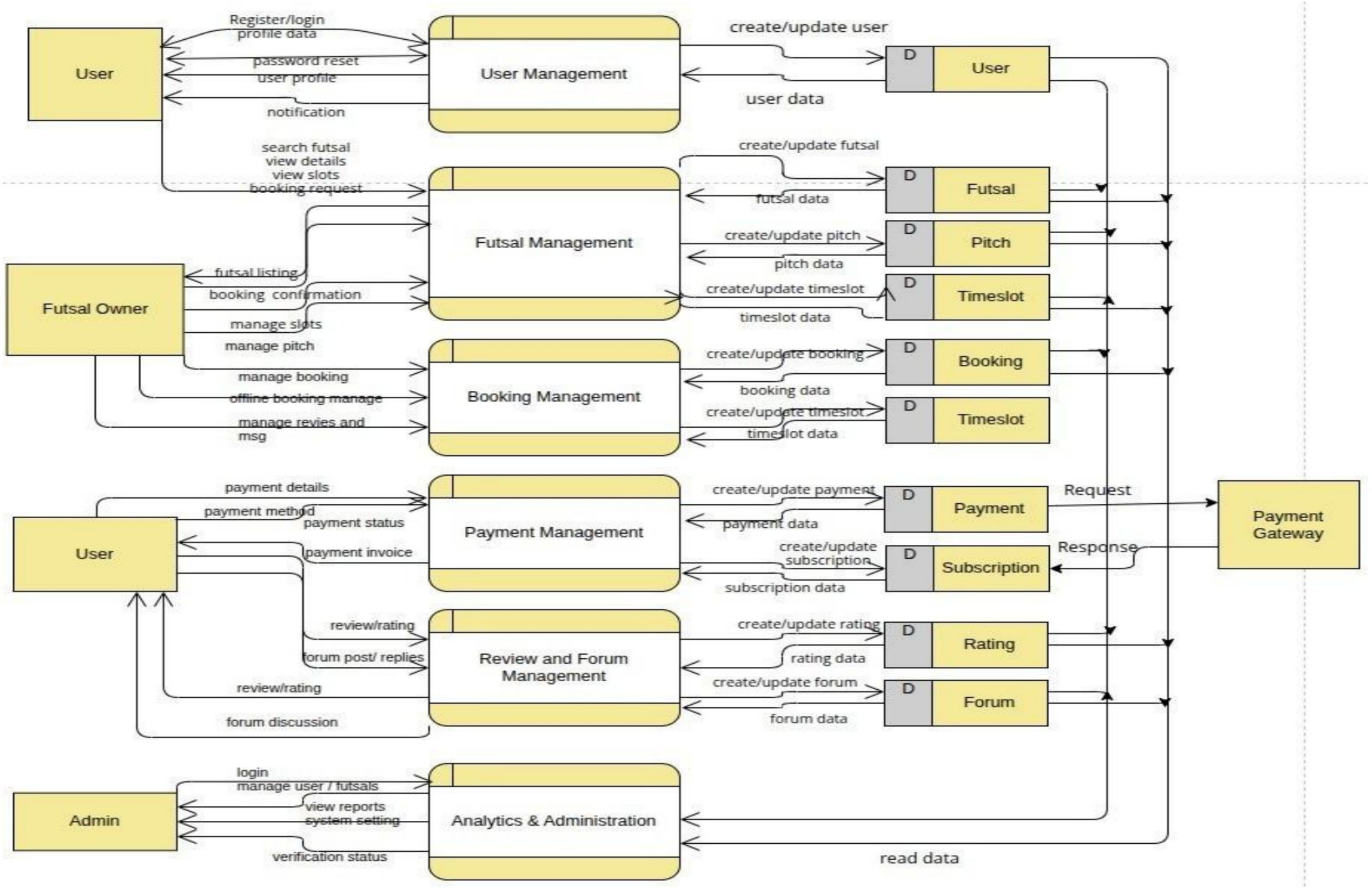
ER Diagram



DFD level -0



DFD level-1



User
Register/login
profile data
password reset
user profile
notification
User Management
create/update user
user data
D
User
search futsal
view details
view slots
booking request
Futsal Management
create/update futsal
futsal data
Futsal
create/update pitch
pitch data
Pitch
create/update timeslot
timeslot data
Timeslot
Futsal Owner
futsal listing
booking confirmation
manage slots
manage pitch
manage booking
offline booking manage
manage revies and msg
Booking Management
create/update booking
booking data
Booking
create/update timeslot
timeslot data
Timeslot
User
payment details
payment method
payment status
payment invoice
Payment Management
create/update payment
payment data
Payment
Request
Payment Gateway
create/update subscription
subscription data
Subscription
Response
review/rating
forum post/ replies
review/rating
forum discussion
Review and Forum Management
create/update rating
rating data
Rating
create/update forum
forum data
Forum
Admin
login
manage user / futsals
view reports
system setting
verification status
Analytics & Administration
read data

Algorithm

A supervised classifier **Logistic Regression** that determines review sentiment. **TF-IDF** feature vectors from review text are passed through the trained model to output a Positive / Negative label.

Why Sentiment Analysis?

To automatically understand whether users are satisfied or dissatisfied with a futsal based on their reviews and use that information to improve recommendations.

Working of Algorithm

The sentiment of each futsal review is classified using Logistic Regression based on the TF-IDF feature

$P(\text{Positive}) = 1 / (1 + e^{-z})$

The weighted score **z** is calculated as:

$z = (w_1 \times x_1) + (w_2 \times x_2) + \ldots + (w_n \times x_n) + b$

Where:

- $x_1, x_2, \ldots, x_n$ = TF-IDF feature values
- $w_1, w_2, \ldots, w_n$ = Learned model weights
- b = Bias/intercept

$P(\text{Positive}) \geq 0.5$ - Positive Review

$P(\text{Positive}) < 0.5$ - Negative Review

The predicted sentiment is then used to identify futsal venues with more positive user feedback and support the recommendation of suitable futsal venues.

Conclusion

Overall,our Futsal Booking and Management System provides a secure, efficient, and scalable platform for managing futsal businesses.

It reduces manual errors and double bookings while streamlining booking, revenue tracking, subscriptions, and customer communication.

Users benefit from real-time court discovery, instant confirmations, verified reviews, and proximity-based recommendations.

The integration of sentiment analysis provides AI-driven insights from user feedback, while the multi-tenant SaaS architecture supports future expansion and nationwide futsal management.

References

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THANK YOU